

Orbital Ordering in 2D Multiferroic VCl_3/HOPG

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Orbital ordering in two-dimensional van der Waals (vdW) materials can give rise to exotic symmetry-broken phases. However, direct experimental evidence of orbital-ordered states has remained largely elusive. Monolayer (ML) VCl_3 , a magnetic vdW material, has been theoretically predicted to host an orbital-ordered phase that breaks the intrinsic C_3 symmetry of the lattice, inducing structural distortion in the chlorine sublattice, and giving rise to ferroelectricity [1]. In this work, we grow ML- VCl_3 on highly oriented pyrolytic graphite (HOPG) using molecular beam epitaxy and investigate its electronic structure using a scanning tunneling microscope (STM). By directly mapping the local density of states (LDOS) of the top-layer chlorine atoms, we visualize the electronic density of the underlying orbital ordering of vanadium atoms. Comparison with theoretical calculations indicates the presence of an antiferro-orbital ordered phase in ML- VCl_3 . Interestingly, we observe that the symmetry-broken ferroelectric ground state can be tuned by an external magnetic field, where the C_2 -symmetric state reverts to the undistorted C_3 symmetry phase beyond a critical field. We further perform a detailed study of ferroelectric domains within the symmetry-broken phase. Our measurements reveal pronounced band bending at ML- VCl_3 island edges and weaker band bending along domain boundaries, consistent with their ferroelectric origin. In addition, we demonstrate local manipulation of these ferroelectric domains via the tip-induced electric field and observe temperature-dependent changes in domain orientation. Together, these findings establish ML- VCl_3 as a field-tunable multiferroic vdW magnet and provide direct experimental evidence that orbital ordering acts as the primary symmetry-breaking mechanism. This work opens pathways toward symmetry-switchable electronics and orbitally engineered quantum functionalities in two-dimensional materials.

[1] Luigi Camerano, *Nano Letters* 25, 4825–4831 (2025).